# Licenced Hospitals, Clinics and Dispensaries in Kenya, Year 2025

**An animated plot of the Kenya population increase.**

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### Introduction

Using the data from Licenced HealthFacilities for the year 2025 [1], we analyze hospital distribution in kenya per county.

[1] Archive - Licenced HealthFacilities for the year 2025

Kenya's healthcare system employs a six-tiered structure. **Level 1** Community Health Units (CHUs) focused on preventive and promotive healthcare care through community health workers. **Level 2** dispensaries and clinics focus on basic outpatient services for common ailments. **Level 3** health centers provide curative and preventative care, including maternal and child health. **Level 4** sub-county and district hospitals offer primary referral services. **Level 5** county referral and teaching hospitals handle secondary referrals and specialized care. Finally, **Level 6** national referral hospitals, such as Kenyatta National Hospital provide the highest level of tertiary care [2] [3].

## Analysis

### Importing Libraries

```
import matplotlib.pyplot as plt
import pygadm
import pandas as pd
import requests
from io import StringIO
import geopandas as gpd
from adjustText import adjust_text
from IPython.display import display, Markdown
import matplotlib.colors as colors
```

### Data

```
# Get the data
health_facilities_data = requests.get(
    "https://kmpdc.go.ke/Registers/H-Facilities.php").text
# Parse the data
health_facilities_df = pd.read_html(StringIO(health_facilities_data))[0]
# Drop columns that are not useful
health_facilities_df = health_facilities_df.drop([
    'View', # Empty column
    'status', # All values are 'Active'
    'Reg_No', # Not useful because it's unique and obscured
], axis=1)
```

[2] Explainer: Six levels of hospitals and services they offer
[3] Healthcare in Kenya

```
# Make levels categorical
health_facilities_df['Level'] = pd.Categorical(
    health_facilities_df['Level'].str.lower().str.strip().str.upper())
# drop duplicates
health_facilities_df = health_facilities_df.drop_duplicates()
health_facilities_df
```

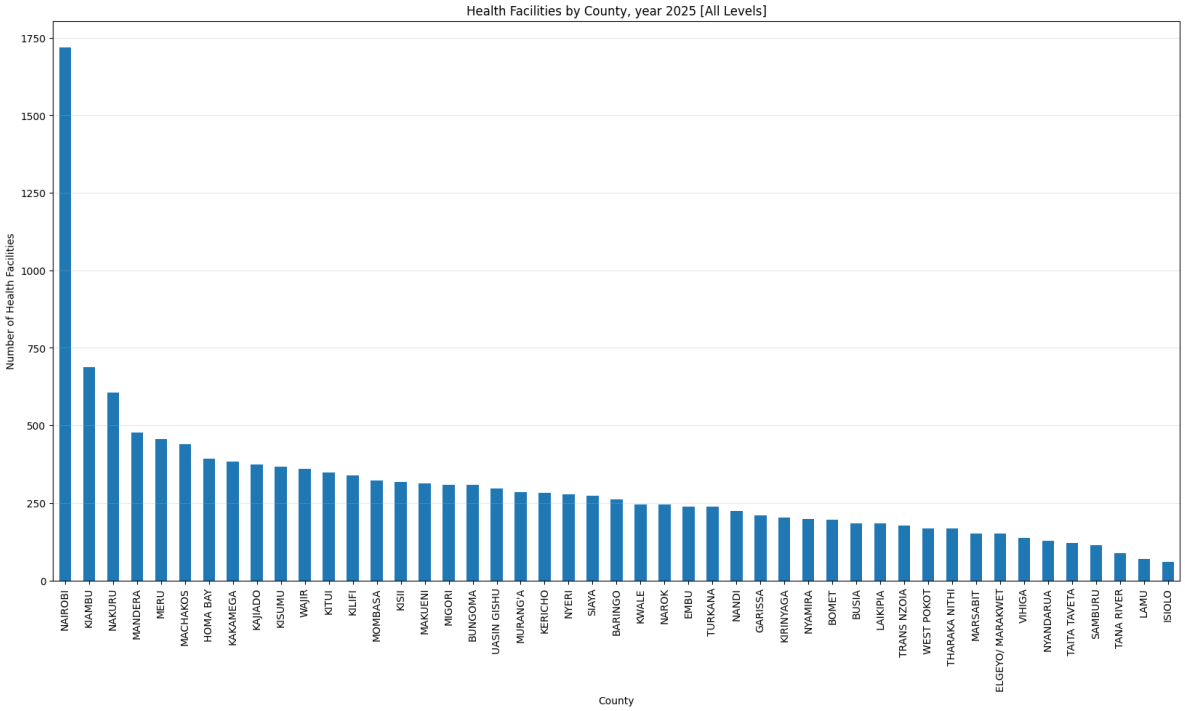
| | Facility_Name | Address | Facility_Typ |
|---|---|---|---|
| 0 | A.C.K ST. PAULS MIHUTI DISPENSARY | P.O BOX 227-10202 KANGEMA | DISPENSAR |
| 1 | NEEMA MEDICARE MEDICAL CENTRE | P.O BOX 59461-00200 NAIROBI | MEDICAL C |
| 2 | AMUNG\'ENTI CATHOLIC DISPESARY | P.O BOX 75-60600 MAUA | DISPENSAR |
| 3 | LIFOG CENTRE MEDICAL CLINIC | P.O BOX 84-80100 MOMBASA | MEDICAL C |
| 4 | RAPHA JOY MEDICAL CLINIC | P.O BOX 244-00221 LAIKIPIA | MEDICAL C |
| ... | ... | ... | ... |
| 14094 | NJORO HUDUMA MEDICAL CLINIC NAKURU | P.O BOX 86-20107 NJORO | NURSING H |
| 14095 | DYNAMED MEDICAL CLINIC | P.O BOX 454-01001 KALIMONI | MEDICAL C |
| 14096 | SLENMARK MEDICAL CLINIC | P.O BOX 267-10304 KUTUS | MEDICAL C |
| 14097 | SLENMAC CARE CENTRE LIMITED | P.O BOX 267- KERUGOYA | MEDICAL C |
| 14098 | LUIZ HOASIS MEDICAL CENTRE EBATE | P.O BOX 366-40200 KISII | MEDICAL C |

```
def health_facilities_county(title: str, levels = None):
    display(Markdown(
        "\n"
        f"### {title}"
        "\n"))
    df = health_facilities_df\
        if levels == None\
        else health_facilities_df[health_facilities_df['Level'].isin(levels)]
    sizes = df.groupby('County').size()
    sizes.sort_values(ascending=False)\
        .plot(kind='bar', figsize=(20, 10))
    plt.title(title)
    plt.ylabel('Number of Health Facilities')
    plt.xlabel('County')
    plt.grid(True, alpha=0.3, axis='y',)
    plt.show()

    print(sizes.describe())

health_facilities_county(
    'Health Facilities by County, year 2025 [All Levels]')
```

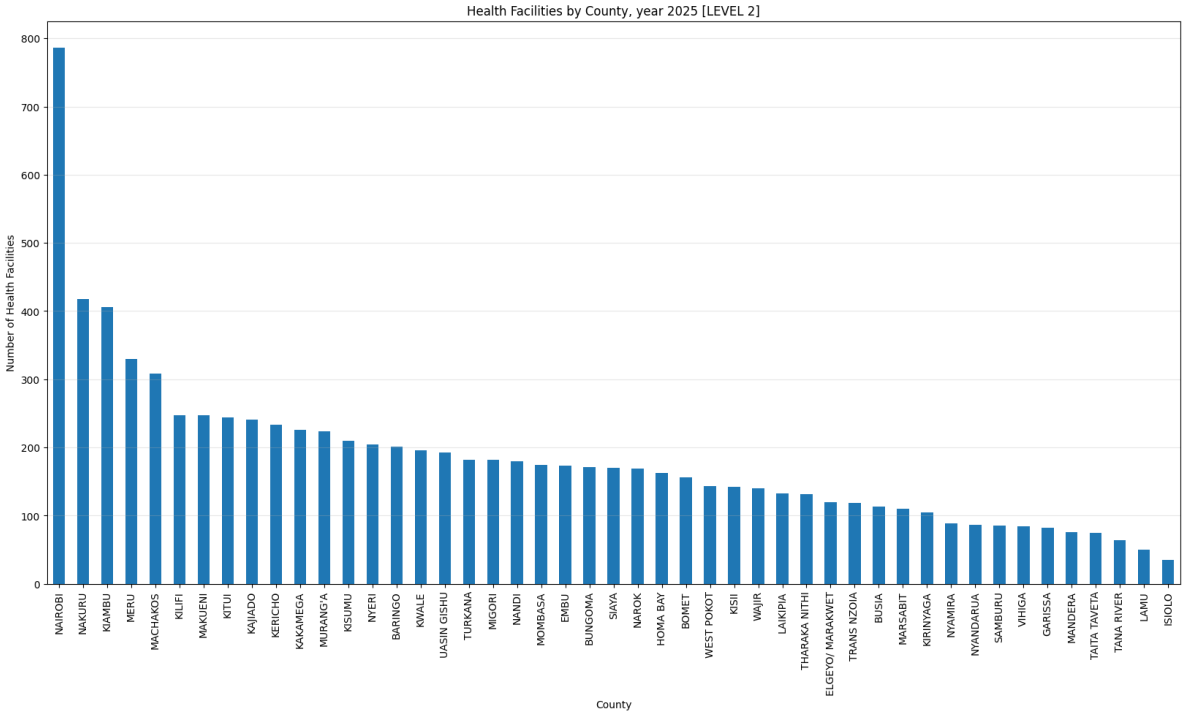
### Health Facilities by County, year 2025 [All Levels]




count 47.000000 mean 299.957447 std 247.828654 min 61.000000 25% 180.500000 50% 261.000000 75% 342.500000 max 1718.000000 dtype: float64

```
health_facilities_county(
    'Health Facilities by County, year 2025 [LEVEL 2]',
    ['LEVEL 2'])
```

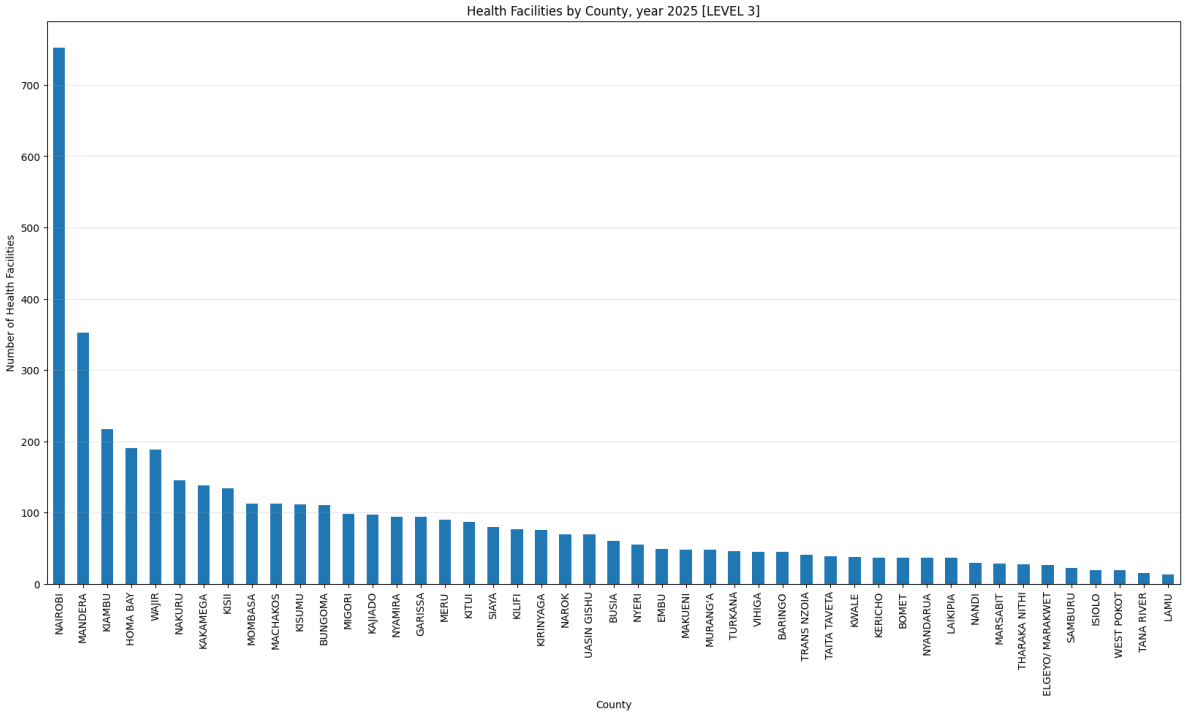
## Health Facilities by County, year 2025 [LEVEL 2]




```
count     47.000000
mean     183.297872
std      122.494350
min       35.000000
25%      111.500000
50%      170.000000
75%      217.000000
max      786.000000
dtype: float64
```

```
health_facilities_county(
    'Health Facilities by County, year 2025 [LEVEL 3]',
    ['LEVEL 3A', 'LEVEL 3B'])
```

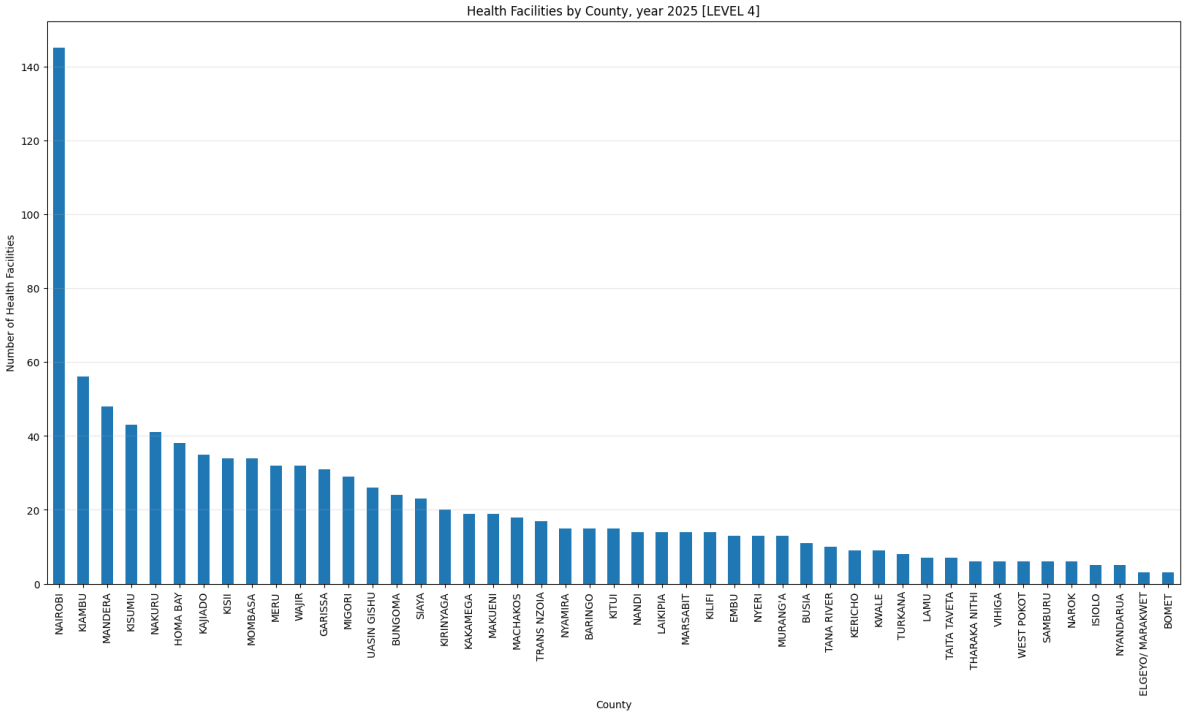
### Health Facilities by County, year 2025 [LEVEL 3]




```
count     47.000000
mean      92.510638
std      116.786850
min       13.000000
25%       36.500000
50%       60.000000
75%      104.000000
max      752.000000
dtype: float64
```

```
health_facilities_county(
    'Health Facilities by County, year 2025 [LEVEL 4]',
    ['LEVEL 4', 'LEVEL 4B'])
```

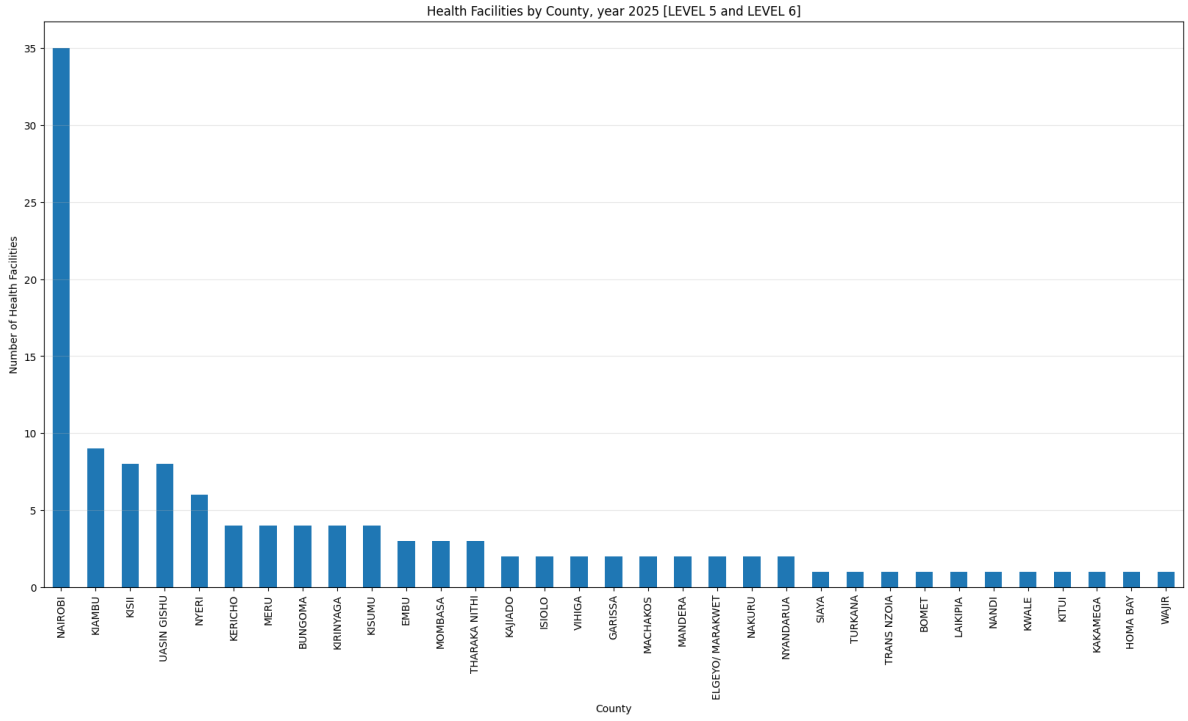
### Health Facilities by County, year 2025 [LEVEL 4]




```
count     47.000000
mean      21.510638
std       22.531980
min        3.000000
25%        8.500000
50%       15.000000
75%       30.000000
max      145.000000
dtype: float64
```

```
health_facilities_county(
    'Health Facilities by County, year 2025 [LEVEL 5 and LEVEL 6]',
    ['LEVEL 5', 'LEVEL 6A', 'LEVEL 6B'])
```

### Health Facilities by County, year 2025 [LEVEL 5 and LEVEL 6]




```
count    33.000000
mean      3.757576
std       6.015762
min       1.000000
25%       1.000000
50%       2.000000
75%       4.000000
max      35.000000
dtype: float64
```

Only three counties have more than 500 helath facilities (Nairobi, Kiambu and Nakuru). Nairobi county leads with 1,718 facilities while Isiolo county tails with 61 facilities.

### Hospital Density

If we gave each health facility a score depending on the level, and sum the scores for each county, we can get a simplistic estimate of the healthcare density in each county. The score is calculated as follows:

```
facility_level_values = {
    j: i
    for i, j
    in enumerate(
        health_facilities_df['Level'].sort_values().unique().tolist(), start = 1)
}
facility_level_values
```

```
{'LEVEL 2': 1,
 'LEVEL 3A': 2,
 'LEVEL 3B': 3,
 'LEVEL 4': 4,
 'LEVEL 4B': 5,
 'LEVEL 5': 6,
 'LEVEL 6A': 7,
 'LEVEL 6B': 8}
```

```
density_data = [
    [
        county,
        int(pd.to_numeric(facilities["Level"].apply(lambda x: facility_level_values[x])).sum
        int(facilities["Level"].count()),
    ]
    for county, facilities
    in health_facilities_df.groupby('County')
]
density_df = pd.DataFrame(density_data, columns=['County', 'Density', 'Count'])\
    .sort_values('Density', ascending=False)\
    .reset_index(drop=True)
density_df
```

| | County | Density | Count |
|---|---|---|---|
| 0 | NAIROBI | 3671 | 1718 |
| 1 | KIAMBU | 1257 | 688 |
| 2 | MANDERA | 1076 | 478 |
| 3 | NAKURU | 949 | 606 |
| 4 | HOMA BAY | 813 | 392 |
| 5 | WAJIR | 706 | 361 |
| 6 | MERU | 698 | 456 |
| 7 | KISUMU | 691 | 368 |

| | County | Density | Count |
|---|---|---|---|
| 8 | MACHAKOS | 672 | 440 |
| 9 | KAKAMEGA | 654 | 384 |
| 10 | KISII | 649 | 318 |
| 11 | MOMBASA | 632 | 323 |
| 12 | KAJIADO | 631 | 375 |
| 13 | BUNGOMA | 599 | 309 |
| 14 | MIGORI | 528 | 309 |
| 15 | UASIN GISHU | 524 | 296 |
| 16 | KITUI | 511 | 347 |
| 17 | KILIFI | 491 | 338 |
| 18 | SIAYA | 452 | 274 |
| 19 | GARISSA | 447 | 209 |
| 20 | MAKUENI | 436 | 314 |
| 21 | NYERI | 433 | 278 |
| 22 | MURANG'A | 395 | 285 |
| 23 | KIRINYAGA | 386 | 204 |
| 24 | KERICHO | 379 | 283 |
| 25 | EMBU | 371 | 238 |
| 26 | BARINGO | 366 | 261 |
| 27 | NYAMIRA | 362 | 198 |
| 28 | NAROK | 356 | 244 |
| 29 | TURKANA | 331 | 237 |
| 30 | KWALE | 328 | 244 |
| 31 | NANDI | 315 | 224 |
| 32 | BUSIA | 305 | 184 |
| 33 | TRANS NZOIA | 292 | 178 |
| 34 | LAIKIPIA | 283 | 183 |
| 35 | BOMET | 258 | 196 |
| 36 | THARAKA NITHI | 236 | 167 |
| 37 | VIHIGA | 231 | 137 |
| 38 | MARSABIT | 227 | 152 |
| 39 | WEST POKOT | 211 | 168 |
| 40 | ELGEYO/ MARAKWET | 201 | 151 |
| 41 | NYANDARUA | 194 | 129 |
| 42 | TAITA TAVETA | 193 | 121 |
| 43 | SAMBURU | 162 | 113 |
| 44 | TANA RIVER | 141 | 89 |
| 45 | LAMU | 113 | 70 |
| 46 | ISIOLO | 111 | 61 |

Get the geometry of the counties in Kenya, GADM (Global Administrative Areas).

```
gdf: gpd.geodataframe.GeoDataFrame = pygadm.Items(
    name="Kenya", content_level=1).rename(columns={"NAME_1": "County"})
gdf = gdf.drop(columns=list(set(gdf.columns) - set(["geometry", "County"])),
               axis=1)
gdf.head(5)
```

| | geometry | County |
|---|---|---|
| 0 | MULTIPOLYGON (((35.7616 -0.1904, 35.7243 -0.19... | Baringo |
| 1 | MULTIPOLYGON (((35.2613 -1.0159, 35.2583 -1.02... | Bomet |
| 2 | MULTIPOLYGON (((34.8778 0.8339, 34.9138 0.8524... | Bungoma |
| 3 | MULTIPOLYGON (((34.0292 -0.0142, 34.0158 -0.02... | Busia |
| 4 | MULTIPOLYGON (((35.5272 0.2167, 35.5261 0.2183... | Elgeyo-Marakwet |

Merge the densities and the geometries

```
def format_county_name(county_name: pd.Series) -> pd.Series:
    return county_name.str.strip().str.lower().str.replace(r'[^a-zA-Z0-9]+', '-', regex=True)
density_df['County'] = format_county_name(density_df['County'])
gdf['County'] = format_county_name(gdf['County'])
merged_gdf = gdf.merge(density_df, on='County', how='left')
merged_gdf.head(5)
```

| | geometry | County | Density | Count |
|---|---|---|---|---|
| 0 | MULTIPOLYGON (((35.7616 -0.1904, 35.7243 -0.19... | Baringo | 366 | 261 |
| 1 | MULTIPOLYGON (((35.2613 -1.0159, 35.2583 -1.02... | Bomet | 258 | 196 |
| 2 | MULTIPOLYGON (((34.8778 0.8339, 34.9138 0.8524... | Bungoma | 599 | 309 |
| 3 | MULTIPOLYGON (((34.0292 -0.0142, 34.0158 -0.02... | Busia | 305 | 184 |
| 4 | MULTIPOLYGON (((35.5272 0.2167, 35.5261 0.2183... | Elgeyo-Marakwet | 201 | 151 |

Plotting the simplistic densities gives us the following map.

```
# Create figure and axis
fig, ax = plt.subplots(1, 1, figsize=(12, 12))

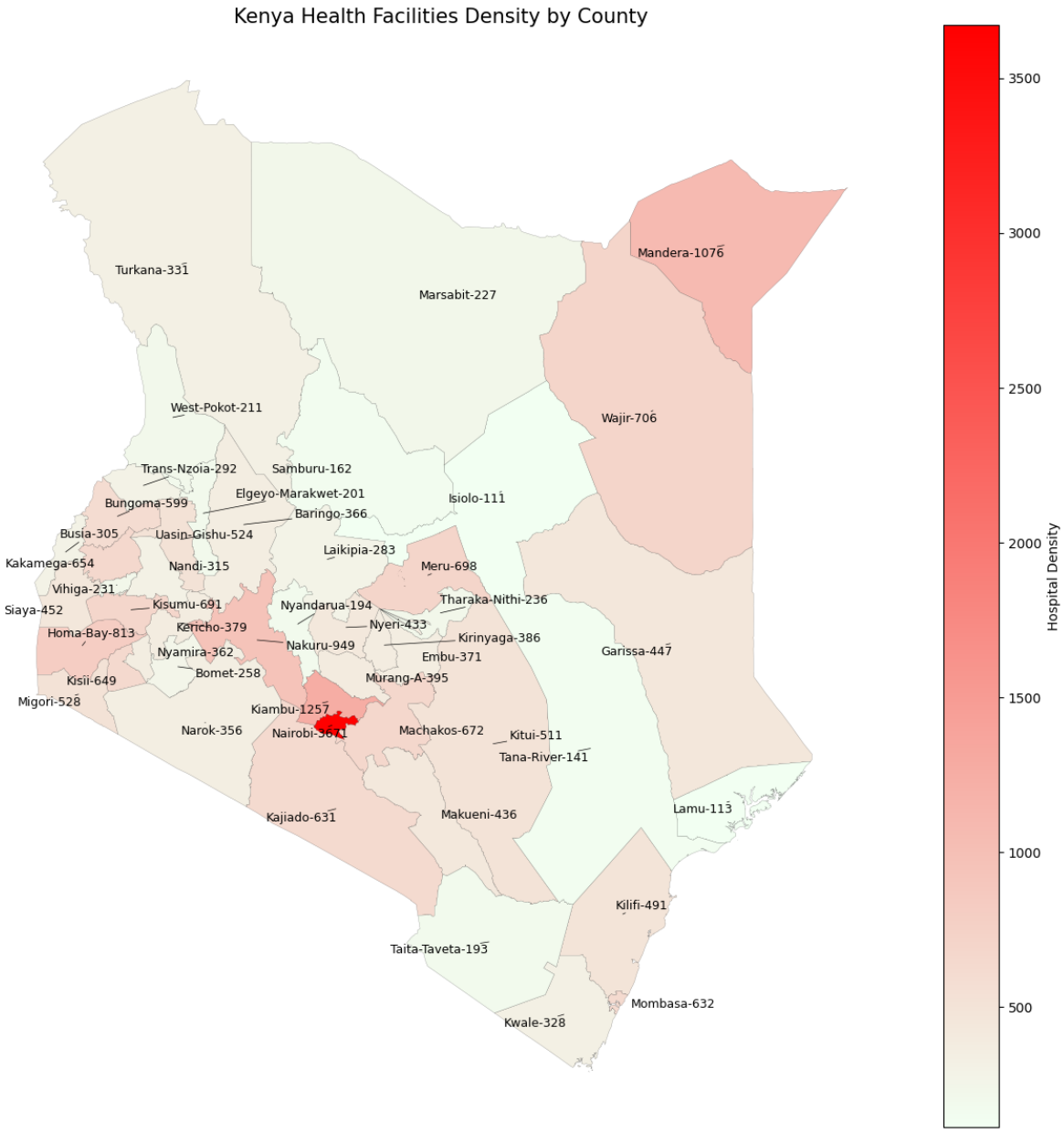
# Create a custom colormap from green to red
cmap = colors.LinearSegmentedColormap.from_list('custom_cmap', ['#f2fff2', '#ff0000'])
```

```
# Plot with hospital count determining the color
merged_gdf.plot(
    ax=ax,
    column='Density',
    cmap=cmap,
    legend=True,
    legend_kwds={'label': "Hospital Density", 'orientation': "vertical"},
    linewidth=0.1, edgecolor='black'
)

texts = [
    ax.text(
        row.geometry.centroid.x,
        row.geometry.centroid.y,
        f"{row['County']}-{row['Density']}",  # Just show the count
        fontsize=9,
        ha='center',
        va='center',
        color='black',
        # fontweight='bold',
        # bbox=dict(facecolor='white', alpha=0.7, boxstyle='round,pad=0.3')
    )
    for idx, row
    in merged_gdf.iterrows()]
adjust_text(texts, arrowprops=dict(arrowstyle='-', color='black', lw=0.5))

# Add title and remove axes
plt.title('Kenya Health Facilities Density by County', fontsize=15)
plt.axis('off')

# Show the map
plt.tight_layout()
plt.show()
```




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